n = 83: inlined field operations, exact changes only (Tier 1 design)

2026-11-04

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Status (2026-11-04): result in. Verdict: improvement; every gate passes and the run is valid `[pcc:unverified]`. R = E2E(A) / E2E(B) = **1.2610 [1.2541, 1.2681]** over 36 alternating pairs, run 41869146827, Tier 1, instance scope. The design sections below are the frozen text, unchanged; they were frozen before any timing on the Tier 1 runner. Every profile figure in them comes from the shared cloud-agent VM; those figures are local diagnostics, used only to rank candidates and size the design. The result section is added with the archived receipt.

Why this run

The n = 83 a = 1 baseline row is the **rank-finder producer** (`koblitz-n83-rankfinder-20261021`, on main, run 39527997793, frozen `rankfinder_source_sha256` `301615b3...`). In that run its end-to-end cold cost was **3.924x** the matched rT770 rho (point estimate; online 0.0933x). That gap is the distance remaining `[pcc:unverified]`. Side B of that run (this run's side A) spent, per process (medians over the run's rows, ms):

stage	ms
root-index build (cold process)	3,231
full scan-order self-test	840
online, one target	193
base load	176
rank stage (rank process)	2,528

So the cold path is setup-bound. A local Callgrind profile of side A's cold process (this VM, `--cache-sim=yes`) puts **21.7%** of all instructions in `Gf2_128::mul` and its callees, and the rank process **25%** `[pcc:unverified]`. The library's product calls `clmul_u64`, which is not `#[inline]` and lives in the library crate, out of line six times per product: about 150 instructions per GF(2^83) product. Round 2's rejected candidate C8 (`koblitz-n83-indexbuild2-20261020`) enabled the carry-less multiply in the producer's state loop, but that cannot inline a non-`#[inline]` function of another crate, so C8 never removed those calls.

1. The change and its proof

Side B's producer is derived from side A's by `make_fieldops_source.py`: counted, exact edits plus one counted call replacement inside `mod wide` (57 products and 24 squarings). `check_inverse()` undoes every edit back to side A's bytes. None of them changes what is computed:

- **F1, inlined field products and squares** (`fast_field`, `fmul`, `fsqr`). The same polynomial product (four carry-less 64 x 64 products; the library's Karatsuba uses three, and both form the one product of the two polynomials) and the library's two folds by the tail t = irr - z^n,

statement for statement. A reduced field element is unique, so every product and square is the library's. Side B is built with `-C target-feature=+pclmulqdq` (`online-b/.cargo/config.toml`). `fast_field::check` runs before any use and fails closed. It checks that the CPU has the instruction, that the field meets the library's fold conditions, and that the inlined product and square equal the library's on edge cases and 65,536 pseudo-random pairs.
- **E1, the state loop's three squarings by the inlined square.** The table map `solver.square` is the GF(2)-linear map with images `gf.sqr(e)` of the unit vectors, so it equals squaring on every element. Its 45 KB of tables leave the state loop.
- **E2, byte-image lookups in every linear map** (`Linear128::apply`). The same eleven lookups of the same tables, the bytes read from `v.to_le_bytes()`.
- **S4, the scan self-test evaluated from per-round tables** (`RoundTables`). `apply` calls `round(right, key)` only with right < 2^half, so table r holds `round(v, keys[r])` for every v < 2^half, filled by `round` itself. The table-evaluated `apply` is `apply` on every index. The self-test makes the parent's three checks of the same function over the same domain in the same order, with the parent's eight-image loop.

The appended test module `fieldops_tests` (run by the workflow's `cargo test --bins`) checks these equalities:

- the inlined product and square against the library on all 2^20 pairs of a 1,024-element sample;
- the square map against the inlined square on 2^18 elements;
- the byte-image lookup against the parent's shift loop for all four maps;
- the table permutation against `apply` on every index of the M38 and M10 domains, for both self-test keys.

2. Candidates and local ranking (diagnostic)

Ranked on this VM: alternating fresh pinned processes of side A and the candidate, the M38 base and point Q0. Layout, index, relation and probe identities were compared on every run.

candidate	local effect (B/A)	status
F1 alone	index build 0.954	in the stack
F1 + E1	index build 0.903	in the stack
F1 + E1 + E2	index build 0.884, rank stage 0.84, online 0.79	in the stack
S4 (microbenchmark of the self-test alone)	0.65	in the stack
nibble tables (5.4 KB per map) instead of E2	index build 0.97	rejected (E2 better)
self-test with the marks bucketed by image	1.2	rejected (slower)
self-test with images computed ahead and their bitmap words prefetched	1.25	rejected (slower)

The frozen stack against side A, from `inputs/local_profile.json`: four alternating cold rounds and three rank rounds. The rank rounds used the first side-B build, whose rank-stage code is the frozen one.

stage	side A (ms)	side B (ms)	B/A
index build	6,575	5,609	0.853
scan self-test	1,410	1,277	0.906
online	685	537	0.784
rank stage	4,750	4,424	0.931

The first side-B build computed the self-test's round count with a runtime division and ran the self-test slower (1.15). The frozen derivation iterates a fixed array of four tables. Every run reproduced side A's stored-state, slot-layout, logical-state and table-map digests, and every rank run reproduced its relation digest, per-attempt probe vector and root calls.

3. Not in scope

The index keys' linear run finder (about 14% of the cold process's instructions, now the largest non-field cost of the build) and the root table's layout are unchanged. So are the probe loops, the rank-stage algorithm, the factor base and the self-test's predicate.

4. Frozen design

The design, pairing, bootstrap, A/A control, Tier 1 records and fail-closed verifier are the parent's (`protocol.json`, `verify_receipt.py`):

- **Panel:** 12 new held-out points (`inputs/cold_targets.json`, disjoint from every earlier n = 83 panel: `inputs/panel_disjointness.json`).
- **Pairs per point:** three alternating primary pairs, two A/A pairs, one planted pair and one tuned-rho process.
- **Rank seeds:** 2101-2136, new on n = 83 (`inputs/seed_disjointness.json`).
- **Primary:** R = sum E2E(A) / sum E2E(B) over the 36 primary pairs, with E2E = rank stage + cold.
- **Decision:** improvement iff the 95% two-stage paired bootstrap lower bound of R exceeds 1 `[pcc:unverified]`.

Arms:

- **Side A** (the row, `online/`, built as landed): kM38RF + cM38RF.
- **Side B** (`online-b/`, +pclmulqdq): kM38FO + cM38FO.
- **A/A:** B against kM38FOaa + cM38FOaa.
- **Planted:** cold pairs cM38FO / cM38FOX. cM38FOX uses `KIC_INDEX_CANON=lift_plant`, so every index key is also checked against lift_scan. Its layout digests are unchanged and its index build is slower.
- **Q0:** run on K600, M10 and M38 by both producers.

The gates are the parent's plus one. Every IC row's summary must name its arm's `field_ops_pipeline`: absent on side A's binary, and `fmul_clmul_inline+sqr_clmul_inline+state_square_clmul+maps_le_bytes+selftest_round_tables` on side B's.

5. Power and runtime

`power_check.py` (`analysis/power.json`) uses the parent's run as the prior and the four local stage factors as the modelled cell. It gives:

quantity	value
expected R at the prior means	1.12
power at a 1.01 effect	0.975
type I error at 1.00	0.0075
A/A point estimates inside [0.97, 1.03]	every time
planted control detected	every time
modelled schedule	about 31 min
modelled job	about 50 min

Local smoke of the frozen schedule

`run_schedule.py --one-per-arm` on this VM ran the first process of every arm in every segment through `arm.py`'s gates: 102 processes, all valid and verified, none with an issue `[pcc:unverified]`. The verifier passes every substantive gate on that partial receipt (`smoke/README.md`, `inputs/local_exactness.json`). The crate tests of `online-b` pass, 38 of 38.

Trigger

`.github/workflows/n83-fieldops.yml`, on a push to `claude/n83-fieldops` restricted by path filters, as the parent's (`workflow_dispatch` is refused for this integration). It clones crypto 770f6a69 from the public `isogeny-labs/crypto`.

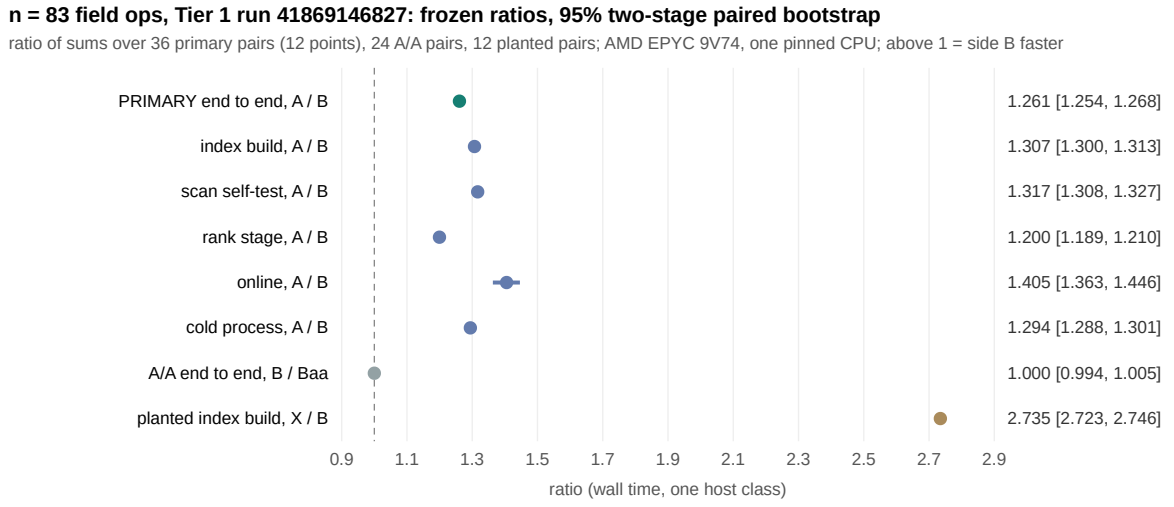
Result

Run 41869146827 (workflow `n83-fieldops.yml`, push trigger, frozen commit `20e1e8db`, attempt 1, conclusion success) on one GitHub-hosted `ubuntu-24.04` runner (`GitHub Actions 1000157034`, image 20261026.354.1, AMD EPYC 9V74, execution CPU 0). The job ran 16:37 to 17:41 UTC (64 min); the schedule took 45 min against the 165-minute guard. The receipt is archived in `run_github_41869146827/` (the 1.5 MB run log in the artifact store, `panel/.artifacts.json`).

Verdict: improvement `[pcc:unverified]`. R = sum E2E(A) / sum E2E(B) = **1.2610 [1.2541, 1.2681]** over 36 alternating pairs (12 points x 3 pairs; two-stage paired bootstrap, 10,000 draws). All 282 scheduled rows ran and are valid, none was skipped by the time guard, and every gate passes:

- one index and layout digest set per base (K600, M10, M38), each equal to the frozen one, and the Q0 identities equal to the archived ones;
- identical relation digests, probe vectors, root calls and counters per rank seed across kM38RF, kM38FO and kM38FOaa, and rank tables equal to the frozen verified tables `[pcc:unverified]`;
- every IC row's `rank_probe_pipeline` and `field_ops_pipeline` equal to its arm's; runner identity, binaries, CPU record and pinning unchanged;
- one scalar per point across every IC and rho process, replayed in the pinned oracle.

`verify_receipt.py` re-run on a copy of the receipt reproduces `panel/result.json` and `panel/result_table.md` byte for byte.



Primary, controls and stages (frozen rule)

stage	side A mean (ms)	side B mean (ms)	A / B [95%]
index build (cold process)	5,802	4,439	1.3069 [1.3001, 1.3134]
scan self-test	1,184	899	1.3165 [1.3077, 1.3269]
rank stage (rank process)	3,790	3,160	1.1996 [1.1894, 1.2099]
online, one target	241	171	1.4053 [1.3632, 1.4460]
cold process	7,528	5,816	1.2944 [1.2877, 1.3011]
end to end (rank + cold)	11,319	8,976	1.2610 [1.2541, 1.2681]

- A/A: B / Baa end to end 0.9998 [0.9944, 1.0051], cold 0.9998 [0.9949, 1.0062], both inside [0.97, 1.03].
- Planted: cM38FOX / cM38FO index build 2.7348 [2.7233, 2.7463], lower bound above 1; the planted processes kept the frozen layout digests.
- Probe law: kM38RF and kM38FO both 0.9789 of the law over 576 attempts, inside the frozen band [0.80, 1.20].
- Order: A first 1.2577, B first 1.2621; every point's ratio lies in [1.250, 1.281].
- Peak RSS and cold minor faults are unchanged (A / B 0.9999 and 1.0001).

The stages moved further than the local model's (0.853, 0.906, 0.931 and 0.784 as B / A): on this host the inlined products shorten the index build by 1.31x, the self-test by 1.32x, the rank stage by 1.20x and the online solve by 1.41x `[pcc:unverified]`.

Distance to rho

In this run's own matched rT770 rho (12 processes, mean cold 2,192 ms), the end-to-end cold gap is **5.164 for side A and 4.096 for side B** (point estimates); S end to end cold is 0.4946 for A and 0.3922 for B in rT770 steps `[pcc:unverified]`. Side A is the code the row landed with, which read 3.924 in its own run (run 39527997793, AMD EPYC 9V45, rT770 step 175.95 ns). Here, on an AMD EPYC 9V74 (rT770 step 247.22 ns), the same code takes 11.32 s end to end against 7.27 s there, while the matched rho's cold mean moved from 1,853 ms to 2,192 ms, so the in-run gap is higher on this host class. That is a cross-run host artifact of the kind `docs/ic/BASELINES.md` already records; only paired deltas within one run are valid, and the paired ratio here is 1.261. Applied to the landed 3.924, it would give about 3.11; that figure is illustrative and not measured.

Limits of this result

- Wall time on one GitHub-hosted runner, one pinned CPU, Tier 1 instance scope, no host-wide exclusivity; not replayed on another host class.
- The gap to rho depends on the host class (above); the paired ratio is the result.
- Factor-base construction is charged to neither arm, as in every n = 83 row, so the cold S is a lower bound.
- The section 5a GAE band is not computable from this run (wall time only).

Files

- **Derivation:** `make_fieldops_source.py` (side B), plus the parents' chain (`make_rankfinder_source.py` and earlier, verbatim).
- **Crates:** `online/` (side A and the tuned rho) and `online-b/` (side B, `.cargo/config.toml`).
- **Experiment scripts:** `restore.py`, `make_targets.py`, `check_panel_disjoint.py`, `check_seed_disjoint.py`, `make_power_prior.py`, `power_check.py`, `make_protocol.py`, `make_schedule.py`, `run_schedule.py`, `arm.py`, `verify_receipt.py` and `test_verify_receipt.py`.
- **Local diagnostics:** `profile/run_cold.sh`, `profile/run_rank.sh`, `profile/raw/` and `inputs/local_profile.json`.
- **Receipt:** `run_github_41869146827/`; figure `figures/result.svg` and `.png` from `make_result_figure.py`; this report as `README.pdf`.